Investigating the effect of passive smoking and cardiovascular disease

Background

Secondhand smoke exposure is defined as non-smokers inhaling the tobacco smoke from smokers within the same environment. This lab aims to study the health effects of secondhand smoke exposure, measured by cotinine, and the risk factor of cardiovascular disease, homocysteine. Exposure to secondhand smoke has been associated with increased risk of inflammation, blood clotting, and endothelial dysfunction, leading to coronary heart disease. Passive smokers have significantly elevated fibrinogen and homocysteine levels (Venn & Britton, 2007). Cigarette smoking increases inflammation, thrombosis, and oxidation of low-density lipoprotein cholesterol. Smoking predisposes the individual to stable angina, acute coronary syndromes, sudden death, and stroke.

Data Description

The National Health and Nutrition Examination Survey NHANES is a program of studies designed to assess the health and nutritional status of adults and children in the United States. We used the National Health and Nutrition Examination Survey NHANES epidemiology dataset from the 2013-2014 cohort. We want to understand whether secondhand smoke exposure contributes to homocysteine even after adjusting for potential confounding variables such as age, sex, race, and BMI. We also want to see what roles these confounding variables play in the relationship between secondhand smoke exposure and heart disease biomarker (homocysteine).

Our primary exposure variable is cotinine, and the primary outcome is homocysteine. We removed missing data from diabetes status, diet quality, and weekly alcoholic drinks. We also separated cotinine into four equal levels: quartile 1, quartile 2, quartile 3, and quartile 4. With each quartile holding 25% of the data of cotinine.

Exploratory Analysis

There were a total of N = 1,535 participants, of those 1,535 participants: N = 368 are within quartile 1, N = 376 are within quartile 2, N = 398 are within quartile 3, and N = 393 are within quartile 4 (Table 1).

There were a total of 1282 (63%) females and 738 (37%) males (Figure 1). Most participants reported their diet quality as good with very good and fair coming second most ranked (Table 1). Age ranges from 20 years old to 85 years old, with the distribution slightly right-skewed (Figure 2). White has the highest population, followed by black and Mexican American, with other Hispanic and other being the lowest (Figure 3). Females have a greater proportion of cotinine exposure than males (Table 1). Seems like most of the homocysteine levels of participants are saturated around 5 and 10(umol/L) with a right-skewed distribution (Figure 4). The BMI distribution has a unimodal distribution with a mean around 25 BMI and a right skewed distribution. It seems like there is an extreme outlier around BMI 140 which is unrealistic (Figure 5). The box plot between race and homocysteine shows that black and white people seem to be the races with the highest homocysteine levels, while Mexican American and Other Hispanic groups seem comparable, except that Mexican American has more outliers who had higher levels of homocysteine (Figure 6). Compared to females, males are more likely to have higher levels of homocysteine. The data is mostly within Cotinine levels of 0 to 2. This is probably because there aren't a lot of people with smoke exposure but various levels of homocysteine. The reasoning for this is probably that there are many other confounders not factored in, therefore the relationship directly between homocysteine and cotinine are unclear. (Figure 7). Interestingly, there are a couple of outliers, one for a large BMI 130 kg/m^2 but relatively low cotinine level at 3 ng/mL and a couple others have a relatively now BMI but high cotinine levels and another couple have a mid range BMI at 45 - 70 (kg/m^2) and cotinine level at 2-6 (ng/mL) (Figure 8). In the scatter plot between cotinine and homocysteine, we can see that most have levels of 0 in terms of cotinine and have a range of values in terms of homocysteine (Figure 9). Most participants fall within 20-40 for the BMI (kg/m^2) and 0- 15 for homocysteine (umol/L) (Figure 10). Overall, most people do not have diabetes and do not drink alcohol (Table 1).

Bivariate Regression Analysis

Simple linear regression was used to model unadjusted association with cotinine (Table 2). Sex male, age, race: Mexican American; other hispanic; other; education: no high school, BMI, alcohol weekly, systolic blood pressure, and total cholesterol were all found to be significant as their p values are less than 0.05. Among the covariates, 6 were found to have a significant positive association with homocysteine. Male had a significant positive association with increased homocysteine levels (β = 1.6467, p = < 2e-16), meaning males have higher levels of homocysteine on average than females. Age also had a significant positive association with increased homocysteine levels (β = 0.090614, p = < 2e-16). The mean homocysteine level increases with age. Those who were under the category of No High School had a significant positive association with increased homocysteine levels (β = 0.5770, p = 0.0159). Therefore, those with no high school education have a higher average homocysteine level. There is significant “unadjusted” association between BMI and homocysteine levels (β = 0.025798, p = 0.00656). As expected, mean homocysteine levels increase with BMI. There is significant “unadjusted” association between the amount of alcohol drank weekly and homocysteine levels (β = 0.21158, p = 2.69e-05). People who drank more alcohol weekly, had a higher average homocysteine level. There is significant “unadjusted” association between the systolic blood pressure and homocysteine levels (β = 0.052833, p = < 2e-16). Those with higher systolic blood pressure had an increased average homocysteine level.

Total cholesterol was found to be significantly negatively associated (β = -0.007751, p = 1.56e-06). As total cholesterol increases, mean homocysteine decreases. Mexican American, other Hispanic, and Other in the race category were found to be significant except for White.

Multiple Regression Analysis

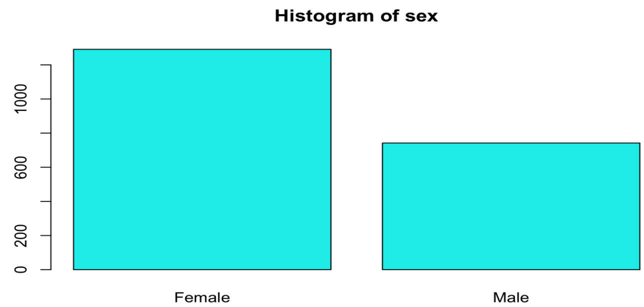
We modeled the adjusted effect of cotinine on homocysteine using a multiple linear regression (Table 2). After controlling for: sex, age, race, education, bmi, diet quality, weekly alcohol drinks, diabetes status, systolic blood pressure, and total cholesterol, the primary exposure variable, cotinine, stayed insignificant as both the simple linear regression and multiple linear regression p- values are larger than the alpha level set at 0.05. This means that after taking these factors into account, cotinine did not make a meaningful difference in the results. The effect size for cotinine from simple linear regression is 0.20108 has decreased to 0.167429 for the multiple linear regression after controlling for all other variables, while the direction of effect has stayed positive. We found that the percent change in the effect size between adjusted and unadjusted is 16.7% which is more than 10% therefore, cotinine was confounded by the inclusion of additional variables.

Other Hispanic, Other Race, Weekly Alcohol Consumption, and systolic blood pressure has become non-significant in the multiple linear regression from simple linear regression.

While White (race), diet quality good, and both have diabetes and no diabetes category became significant from simple linear regression to multiple regression.

Male, age, Mexican American, BMI, total cholesterol significance has stayed the same. This means that these variables have significant influence on homocysteine.

White(race), No high school, high school diploma, Poor Diet Quality, and No Diabetes have direction of change in their respective effect size. It's counter intuitive that No high school and High school diploma has an inverse relationship with homocysteine. After looking at their respective significance, variables No high school and High school diploma are not statistically significant at alpha = 0.05, meaning they do not have significant influence on homocysteine.



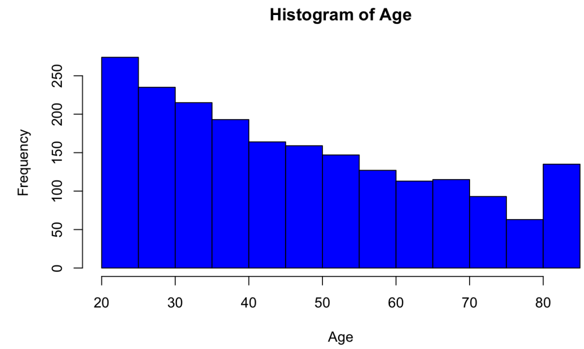
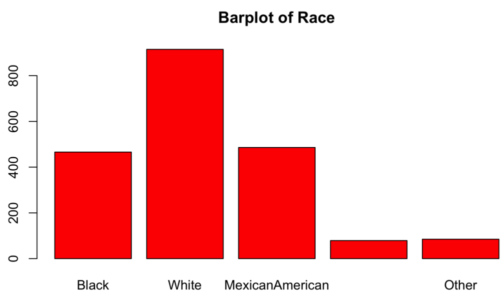


Figure 1: Female v male distribution

Figure 2: Histogram of Age



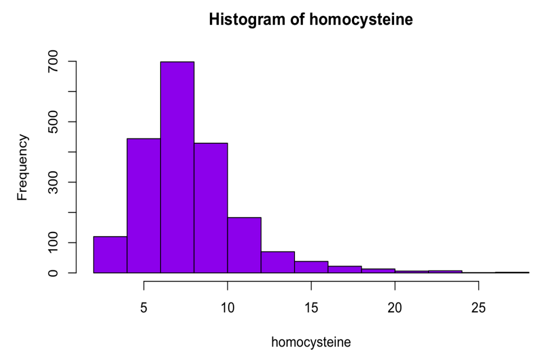


Figure 3: Barplot of Race

Figure 4: Histogram of Homocysteine

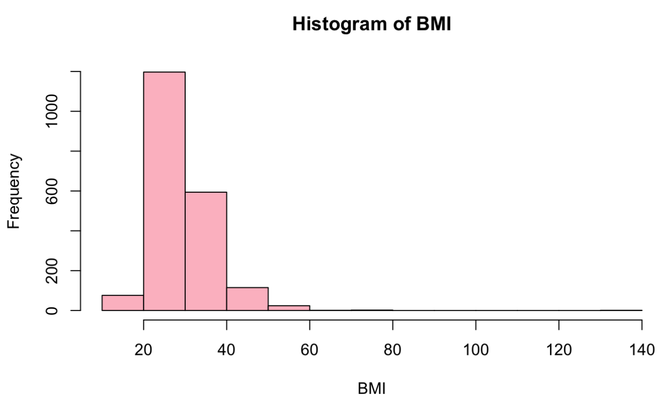


Figure 5: Histogram of BMI

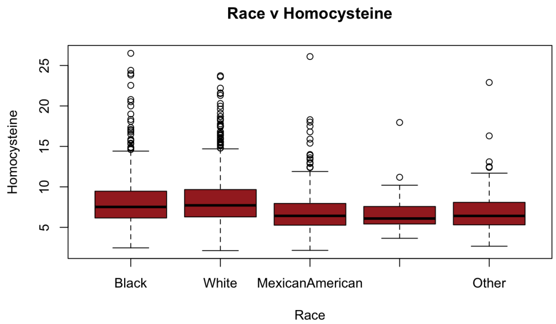


Figure 6: Side by side boxplot Race and Homocysteine

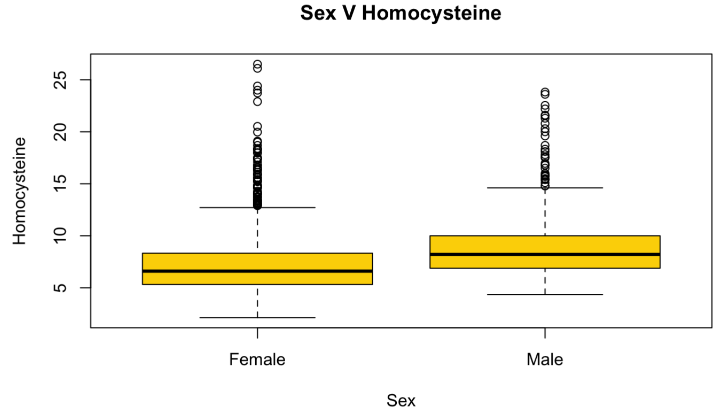


Figure 7: Boxplot between sex and homocysteine

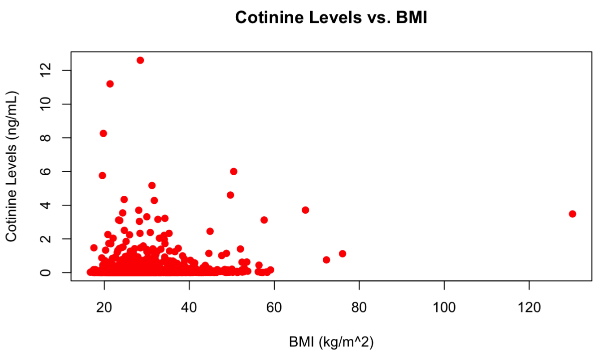


Figure 8: Scatterplot between Cotinine levels and BMI

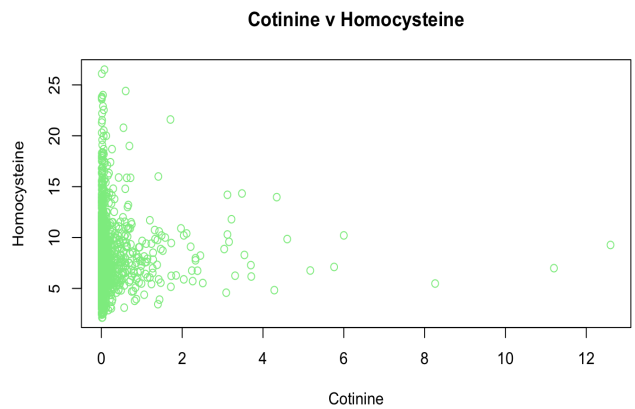


Figure 9: Scatterplot between cotinine and homocysteine

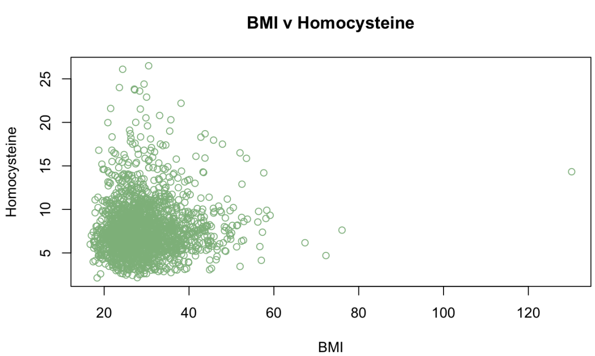


Figure 10: Scatterplot between BMI and Homocysteine

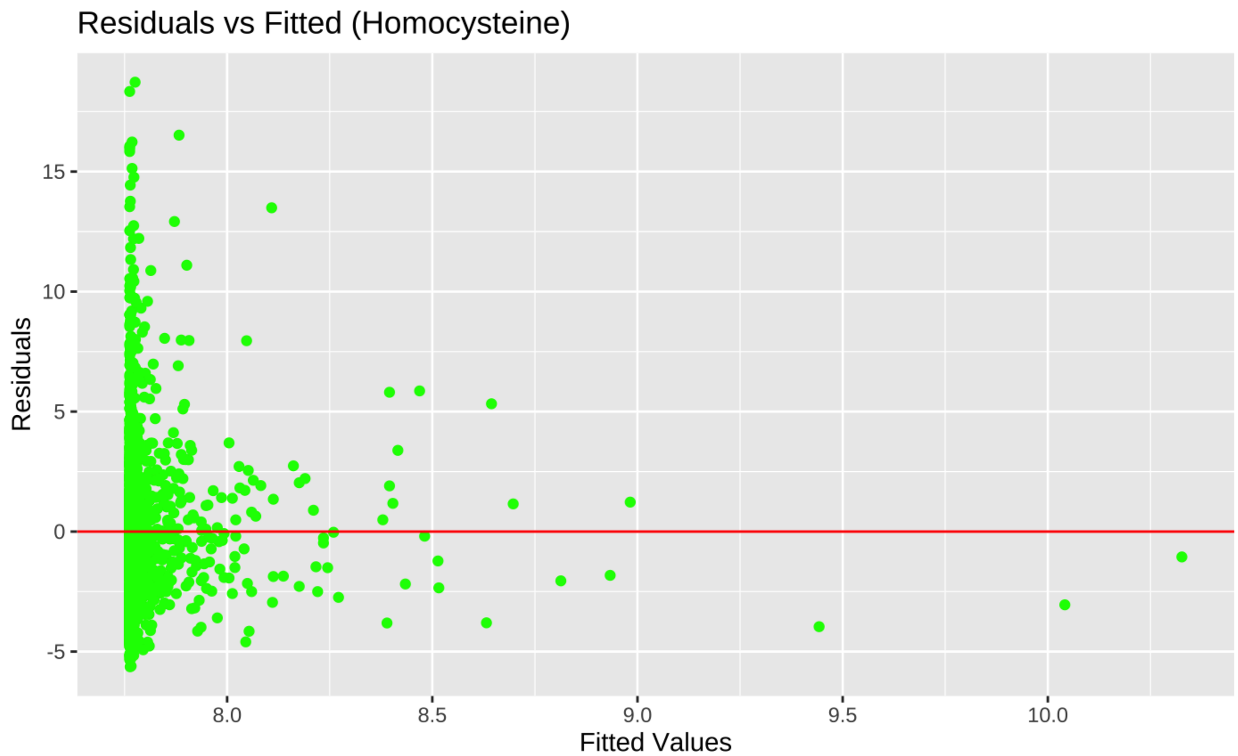


Figure 11: Residual plot for the simple linear regression of Cotinine versus Homocysteine. The linearity assumption does not seem to be met because there are signs of heteroskedasticity. The spread of residuals seem to increase with the predicted Homocysteine values therefore we observe a non-linearity pattern. This suggests that there might be confounders affecting cotinine besides homocysteine.

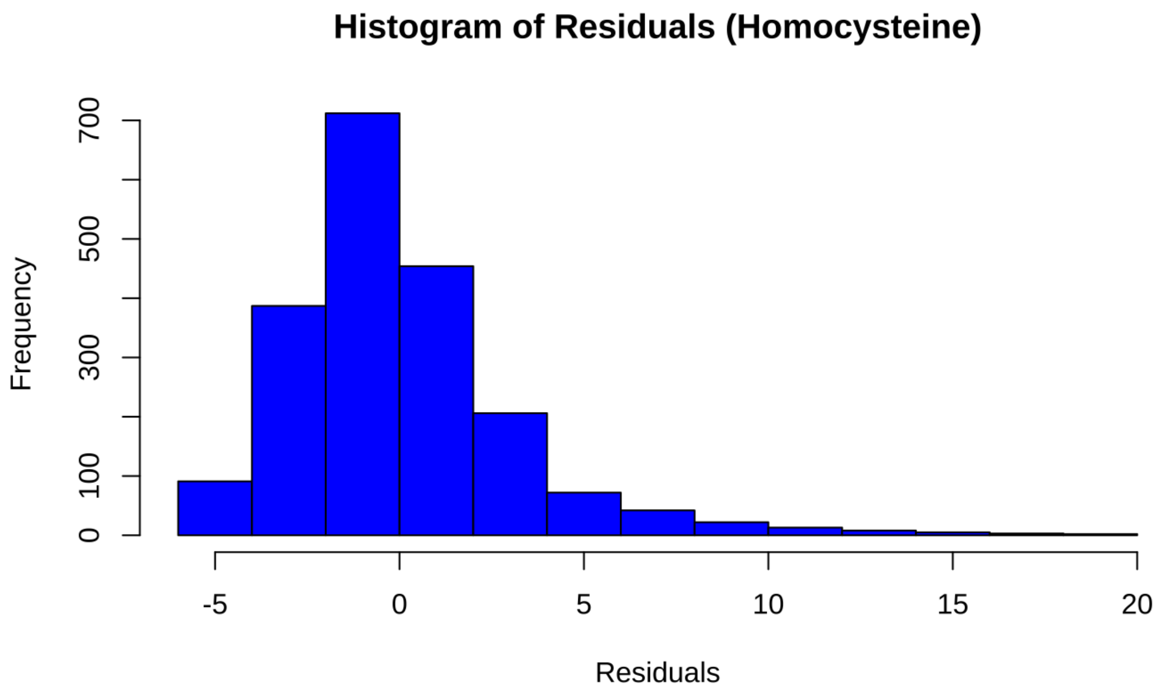


Figure 12: Histogram of residuals from simple linear regression model of Homocysteine versus Cotinine. This plot is used to assess normality of residuals assumption. The graph is unimodal and near symmetric with right skewed.

Table 1: Descriptive statistics for Analyzed cohort. Continuous variables are summarized with mean and standard deviation in parentheses. The proportions are given for each level of categorical variables with count and percentages in parentheses. P values are significant at alpha less than equal to 0.05.

Variable	**Overall** N = 1,533[1]	**quartile 1** N = 368[1]	**quartile 2** N = 375[1]	**quartile 3** N = 398[1]	**quartile 4** N = 392[1]	**p-value**[2]
homocysteine	7.79 (2.91)	7.20 (2.74)	7.92 (3.00)	7.94 (3.00)	8.05 (2.83)	<0.001
sex						<0.001

Female	891 / 1,533 (58%)	265 / 368 (72%)	218 / 375 (58%)	209 / 398 (53%)	199 / 392 (51%)	
Male	642 / 1,533 (42%)	103 / 368 (28%)	157 / 375 (42%)	189 / 398 (47%)	193 / 392 (49%)	
Age in years	46.18 (17.80)	47.07 (18.34)	47.69 (17.46)	46.68 (17.74)	43.38 (17.43)	0.003
race						<0.001
MexicanAmerican	361 / 1,533 (24%)	114 / 368 (31%)	95 / 375 (25%)	75 / 398 (19%)	77 / 392 (20%)	
OtherHispanic	57 / 1,533 (3.7%)	13 / 368 (3.5%)	17 / 375 (4.5%)	14 / 398 (3.5%)	13 / 392 (3.3%)	
White	725 / 1,533 (47%)	196 / 368 (53%)	184 / 375 (49%)	190 / 398 (48%)	155 / 392 (40%)	
Black	332 / 1,533 (22%)	38 / 368 (10%)	61 / 375 (16%)	97 / 398 (24%)	136 / 392 (35%)	
Other	58 / 1,533 (3.8%)	7 / 368 (1.9%)	18 / 375 (4.8%)	22 / 398 (5.5%)	11 / 392 (2.8%)	
education						<0.001
NoHighSchool	163 / 1,533 (11%)	37 / 368 (10%)	46 / 375 (12%)	45 / 398 (11%)	35 / 392 (8.9%)	

SomeHighSchool	180 / 1,533 (12%)	34 / 368 (9.2%)	41 / 375 (11%)	42 / 398 (11%)	63 / 392 (16%)	
HighSchool	333 / 1,533 (22%)	66 / 368 (18%)	78 / 375 (21%)	76 / 398 (19%)	113 / 392 (29%)	
SomeCollege	443 / 1,533 (29%)	108 / 368 (29%)	92 / 375 (25%)	121 / 398 (30%)	122 / 392 (31%)	
College	414 / 1,533 (27%)	123 / 368 (33%)	118 / 375 (31%)	114 / 398 (29%)	59 / 392 (15%)	
BMI	29.32 (7.19)	27.90 (5.87)	29.23 (6.38)	28.99 (6.22)	31.07 (9.33)	<0.001
diet_quality						0.001
Excellent	124 / 1,532 (8.1%)	32 / 368 (8.7%)	27 / 374 (7.2%)	36 / 398 (9.0%)	29 / 392 (7.4%)	
VeryGood	363 / 1,532 (24%)	96 / 368 (26%)	99 / 374 (26%)	93 / 398 (23%)	75 / 392 (19%)	
Good	612 / 1,532 (40%)	157 / 368 (43%)	157 / 374 (42%)	160 / 398 (40%)	138 / 392 (35%)	
Fair	360 / 1,532 (23%)	72 / 368 (20%)	77 / 374 (21%)	90 / 398 (23%)	121 / 392 (31%)	
Poor	73 / 1,532 (4.8%)	11 / 368 (3.0%)	14 / 374 (3.7%)	19 / 398 (4.8%)	29 / 392 (7.4%)	

Number of alcoholic drinks per week						0.015
0	1,079 / 1,533 (70%)	274 / 368 (74%)	269 / 375 (72%)	277 / 398 (70%)	259 / 392 (66%)	
1	200 / 1,533 (13%)	48 / 368 (13%)	51 / 375 (14%)	52 / 398 (13%)	49 / 392 (13%)	
2	116 / 1,533 (7.6%)	18 / 368 (4.9%)	24 / 375 (6.4%)	25 / 398 (6.3%)	49 / 392 (13%)	
3	49 / 1,533 (3.2%)	4 / 368 (1.1%)	11 / 375 (2.9%)	20 / 398 (5.0%)	14 / 392 (3.6%)	
4	23 / 1,533 (1.5%)	6 / 368 (1.6%)	8 / 375 (2.1%)	6 / 398 (1.5%)	3 / 392 (0.8%)	
5	19 / 1,533 (1.2%)	5 / 368 (1.4%)	3 / 375 (0.8%)	7 / 398 (1.8%)	4 / 392 (1.0%)	
6	11 / 1,533 (0.7%)	2 / 368 (0.5%)	3 / 375 (0.8%)	4 / 398 (1.0%)	2 / 392 (0.5%)	
7	36 / 1,533 (2.3%)	11 / 368 (3.0%)	6 / 375 (1.6%)	7 / 398 (1.8%)	12 / 392 (3.1%)	
diabetes_status						0.7
Borderline	28 / 1,533 (1.8%)	10 / 368 (2.7%)	5 / 375 (1.3%)	6 / 398 (1.5%)	7 / 392 (1.8%)	

No	1,371 / 1,533 (89%)	324 / 368 (88%)	336 / 375 (90%)	354 / 398 (89%)	357 / 392 (91%)	
Yes	134 / 1,533 (8.7%)	34 / 368 (9.2%)	34 / 375 (9.1%)	38 / 398 (9.5%)	28 / 392 (7.1%)	
Systolic blood pressure (mm Hg)	123.11 (19.33)	122.12 (21.15)	123.15 (18.96)	122.62 (18.51)	124.50 (18.68)	0.4
Total cholesterol (mg/dL)	200.50 (42.26)	206.71 (44.99)	204.45 (41.50)	198.06 (41.03)	193.39 (40.41)	<0.001

[1] Mean (SD); n / N (%)

[2] One-way analysis of means (not assuming equal variances); Pearson’s Chi-squared test

Table 2: Summary of adjusted and unadjusted association with cotinine

Results for unadjusted associations are based on similar linear regression with cotinine as the outcome and the indicated variable as the single variable in the model. Results for adjusted associations are based on a single multiple regression model with the indicated variables included in the model.

	Unadjusted Associations		Adjusted Associations	
Variable	**Effect Size (Slope)**	**p-value**	**Effect Size (Slope)**	**p-value**
Cotinine	0.20108	0.074	0.167429	0.089256
female	<ref>	<ref>	<ref>	<ref>
sexMale	1.6467	<2e-16 **	1.477266	< 2e-16 ***
Age	0.090614	<2e-16 **	0.081041	< 2e-16 ***
Race				
Mexican American	-1.40751	1.26e-12 *	-0.796496	0.000155 ***
Other Hispanic	-1.67518	6.29e-06 *	-0.541540	0.134974
White	0.03248	0.85111	-0.457266	0.008709 *
Black	<ref>	<ref>	<ref>	<ref>
Other	-1.02928	0.00396 *	-0.661034	0.072107
Highest Education level (%)				
No High School	0.5770	0.0159 **	-0.129358	0.626543
Some High School (no diploma)	0.2689	0.2541	0.127111	0.597167
High School (diploma)	0.3613	0.0748	-0.072519	0.702975

Some College (no degree)	-0.22597	0.2319	-0.098208	0.572908
College (degree)	<ref>	<ref>	<ref>	<ref>
BMI	0.025798	0.00656 **	0.019182	0.047467 *
Diet quality				
Excellent	<ref>	<ref>	<ref>	<ref>
Very Good	0.135365	0.615	0.342168	0.188159
Good	0.03226	0.899	0.691894	0.005266 **
Fair	-0.13402	0.620	0.433735	0.103101
Poor	-0.29997	0.433	0.447293	0.240601
Alcohol Weekly	0.21158	2.69e-05 **	0.053135	0.227715
Diabetes Status				
Boarderline	<ref>	<ref>	<ref>	<ref>
diabetes_statusNo	-0.5833	0.2740	1.275831	0.015189 *
diabetes_statusYes	0.9450	0.0988	1.303396	0.021318 *
Systolic Blood Pressure	0.052833	< 2e-16 **	0.002019	0.605970
Total Cholesterol	-0.007751	1.56e-06 *	-0.006227	4.62e-05 ***